

BINDKEY – RISK ASSESSMENT MATRIX

1. BindKey Firmware Compromise

Responsible: William

Status: IN PROGRESS

A malicious or altered firmware could be installed on the BindKey, allowing security mechanisms to be bypassed.

- Prevention: Use only verified and approved firmware versions
- Detection: Check firmware authenticity at device startup
- Correction: Reinstall a trusted firmware version

2. Physical Attack on the Secure Element

Responsible: Pierre-Louis

Status: IN PROGRESS

An attacker could physically tamper with the device to try to extract sensitive cryptographic keys.

- Prevention: Physically protect the device and critical components
- Detection: Detect signs of physical tampering
- Correction: Automatically block access to stored keys

3. Biometric Data Leakage

Responsible: Pierre-Louis

Status: IN PROGRESS

Sensitive biometric data could be exposed, leading to privacy and regulatory issues.

- Prevention: Store biometric data only inside the BindKey
- Detection: Monitor access to biometric information
- Correction: Reset biometric data and notify the user

4. Server Compromise

Responsible: Jassime & Marwa

Status: IN PROGRESS

The central server could be compromised, exposing user information, access rights, or recovery keys.

- Prevention: Restrict access and protect stored data
- Detection: Monitor abnormal access attempts
- Correction: Restore the server and rotate compromised keys

5. Fake Enroller Creating Unauthorized BindKeys

Responsible: Jean-Baptiste

Status: IN PROGRESS

An unauthorized person could impersonate an enroller and create fraudulent BindKeys.

- Prevention: Secure access to enroller features
- Detection: Detect unusual enrollment activities
- Correction: Block the enroller account and revoke affected BindKeys

6. Loss or Corruption of Volume Encryption Keys

Responsible: Jassime & Marwa

Status: IN PROGRESS

Encryption keys associated with a volume could be lost or corrupted, making data inaccessible.

- Prevention: Securely back up encryption keys
- Detection: Verify that volumes can be correctly opened
- Correction: Generate new keys when recovery is possible

7. API Attack (Injection or Token Theft)

Responsible: Jassime

Status: IN PROGRESS

Attackers could exploit the communication interfaces to gain unauthorized access.

- Prevention: Validate all incoming data
- Detection: Monitor abnormal behavior and requests
- Correction: Invalidate access tokens and fix vulnerabilities

8. Unauthorized BindKey Reset

Responsible: Jean-Baptiste

Status: IN PROGRESS

A BindKey could be reset without proper authorization, resulting in loss of control or data.

- Prevention: Require identity verification or a recovery code
- Detection: Monitor reset attempts
- Correction: Cancel the reset and block the device

9. Volume Key Sent to the Wrong User

Responsible: Marwa

Status: IN PROGRESS

During volume sharing, a key could be mistakenly sent to an unauthorized user.

- Prevention: Verify user identity and permissions before sharing
- Detection: Review assigned access rights
- Correction: Cancel the sharing and change the encryption key

10. Offline Mode Bypass

Responsible: William

Status: IN PROGRESS

A BindKey could remain usable offline for longer than intended, especially after being lost or stolen.

- Prevention: Limit the allowed offline usage duration
- Detection: Detect long periods without synchronization
- Correction: Block the device until synchronization

11. Malicious USB Device Connected to the BindKey

Responsible: William

Status: IN PROGRESS

A malicious USB storage device could disrupt or compromise the BindKey.

- Prevention: Verify that connected USB devices are compliant

- Detection: Detect abnormal device behavior
- Correction: Reject the USB device

12. Incorrect Time Synchronization

Responsible: Jean-Baptiste & William

Status: IN PROGRESS

Incorrect system time can affect offline access duration and security checks.

- Prevention: Synchronize time during server connections
- Detection: Detect inconsistent timestamps
- Correction: Automatically correct the system time

13. Server Overload

Responsible: Jassime

Status: IN PROGRESS

High usage could overload the server and degrade system availability.

- Prevention: Properly size the server infrastructure
- Detection: Monitor response times
- Correction: Optimize or reconfigure the server

14. Biometric Authentication Failure or Bypass

Responsible: Pierre-Louis

Status: IN PROGRESS

The biometric system may fail to recognize a legitimate user or be bypassed.

- Prevention: Properly configure biometric parameters
- Detection: Detect repeated authentication failures
- Correction: Re-enroll biometric data

15. Information Leakage During Key Sharing

Responsible: Marwa

Status: IN PROGRESS

Sensitive information could be exposed when encryption keys are shared between users.

- Prevention: Restrict and control key sharing
- Detection: Audit key transfer operations
- Correction: Revoke shared keys

16. Slow Encryption Performance on Large Files

Responsible: William

Status: IN PROGRESS

Encrypting large files may significantly slow down data transfers.

- Prevention: Optimize overall system performance
- Detection: Regular performance testing
- Correction: Improve firmware efficiency

17. Delayed Log Synchronization

Responsible: Jassime

Status: IN PROGRESS

Logs may not be synchronized in time, reducing traceability in case of incidents.

- Prevention: Schedule regular log synchronization
- Detection: Detect synchronization delays
- Correction: Force manual or automatic synchronization

RISK ASSESSMENT MATRIX				
Impact Probability	LOW	MODERATE	MAJOR	CRITICAL
VERY LIKELY			#5	
LIKELY	#17		#7	#1,#4,#8,#10, #11
UNLIKELY		#12,#13	#6,#9,#15	#2,#3,#14
VERY UNLIKELY	#16			